DATA STRUCTURES

LAB NO. 12

TASK NO. 01

NAME : ALEENA AKHTAR

SAP ID : 69273

SUBMITTED TO : MISS NEHAN SAHAR

DATE : 2ND MAY, 2026

Riphah International University

Gulberg Green Campus

‘TREE’

Task 1: Identify Tree Properties

```
      A
     / \
    B   C
   / \   \
  D   E   F
```

1. Root Node:

The root node is the topmost node of the tree.
Root Node = A

2. Leaf Nodes:

Leaf nodes are the nodes that have no children.
Leaf Nodes = D, E, F

3. Degree of Node B:

The degree of a node is the number of children it has.
Node B has two children (D and E)
Degree of B = 2

4. Height of the Tree:

Height is the number of edges in the longest path from root to a leaf node.

Longest path:

- A → B → D (or A → B → E or A → C → F)

Number of edges = 2

Height of Tree = 2

5. Level of Node E:

Level is counted from the root:

- Level 0 → A
- Level 1 → B, C
- Level 2 → D, E, F

Level of Node E = 2

THANK YOU!!